

# Assignment 1 — Pima Indians Diabetes

## EDA & CLASSIFICATION REPORT

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**Dataset shape: 768 rows × 9 columns**

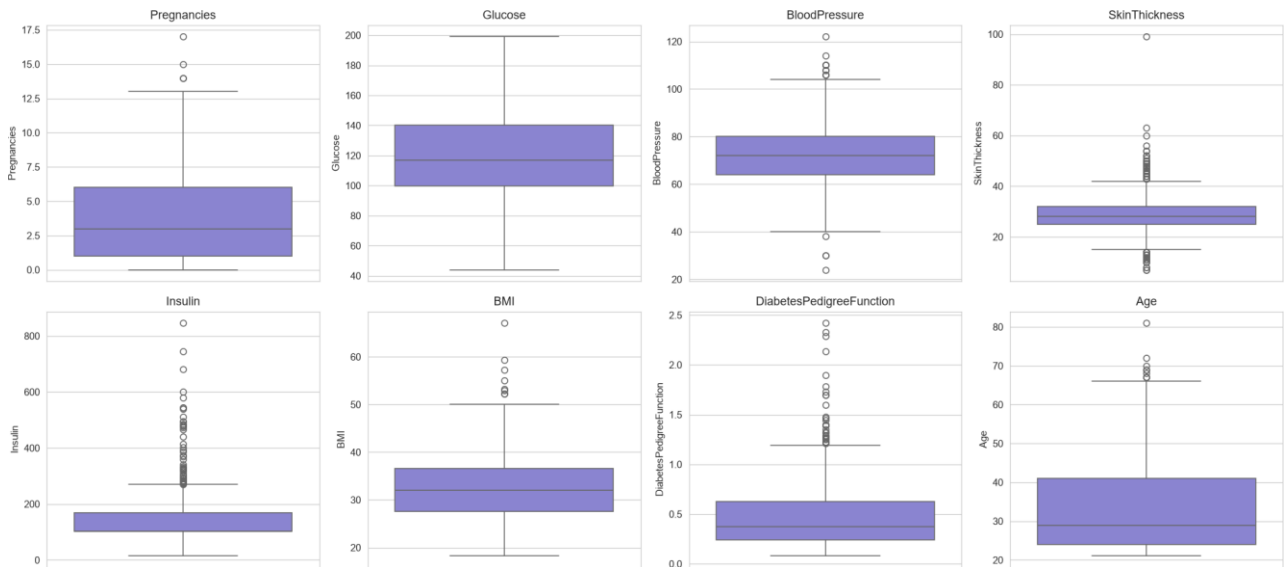
## 1. Data Quality Report

Missing values were imputed using class-conditional medians before this report was generated.

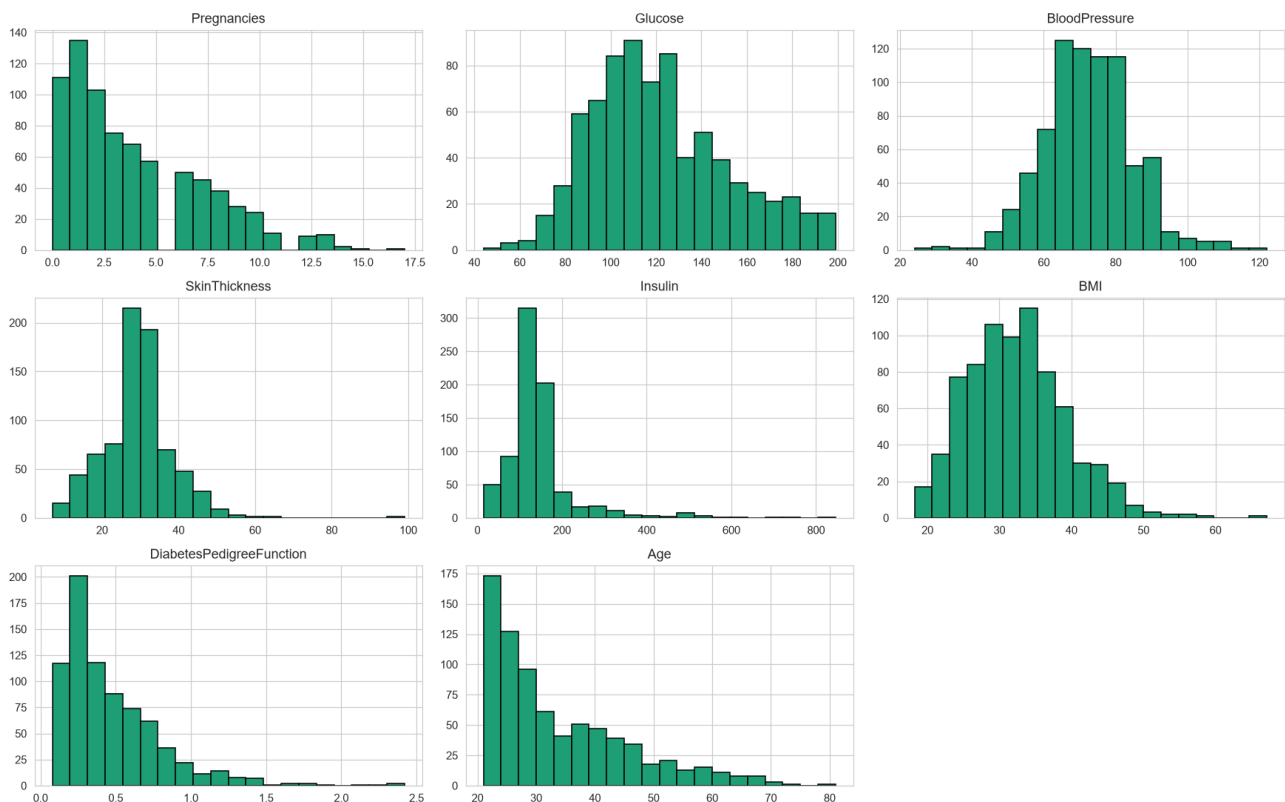
|                            | dtype   | cardinality |
|----------------------------|---------|-------------|
| Pregnancies                | int64   | 17          |
| Glucose                    | float64 | 135         |
| Blood Pressure             | float64 | 47          |
| Skin Thickness             | float64 | 50          |
| Insulin                    | float64 | 187         |
| BMI                        | float64 | 247         |
| Diabetes Pedigree Function | float64 | 517         |
| Age                        | int64   | 52          |
| Outcome                    | int64   | 2           |

## 2. Visualizations

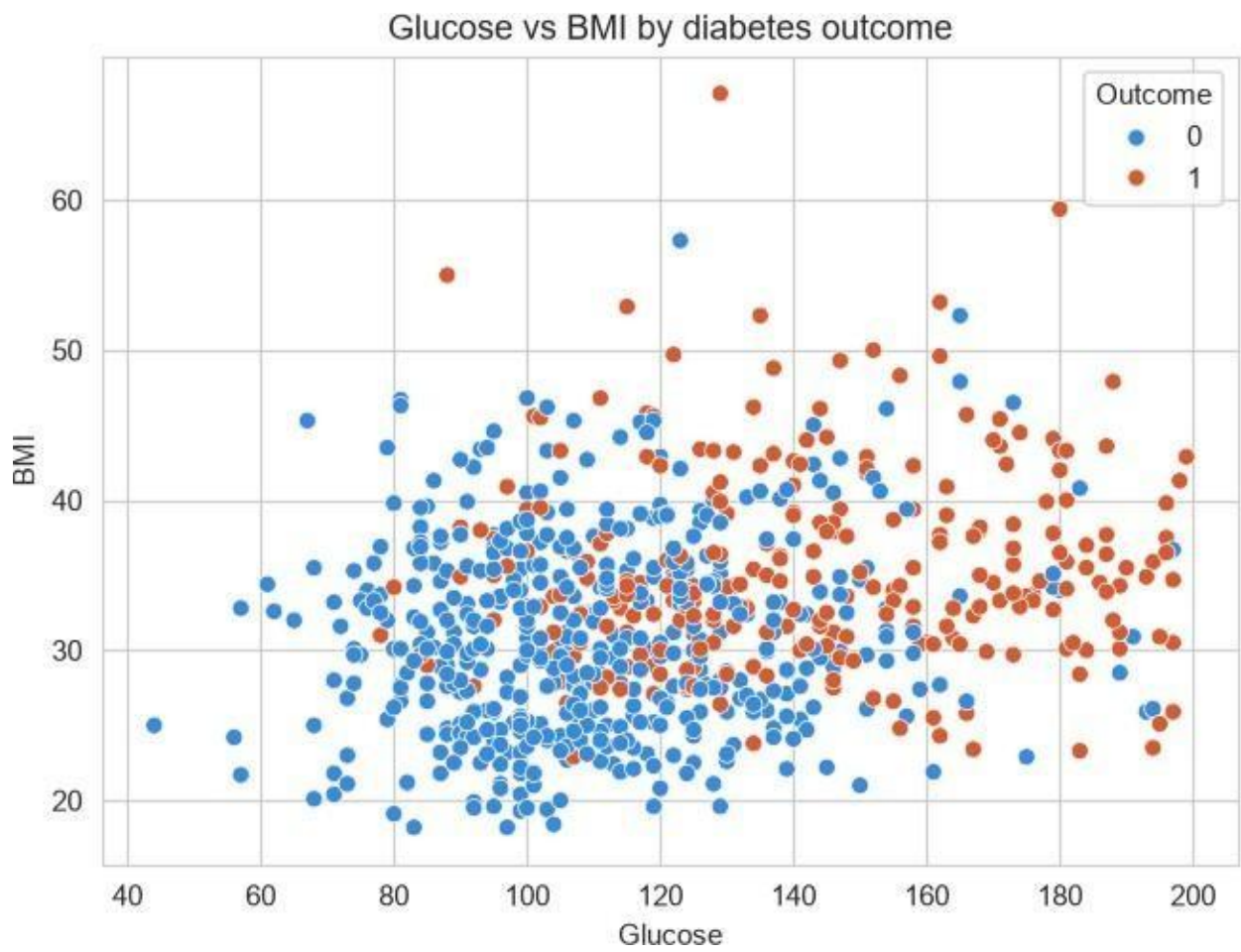
### Boxplots



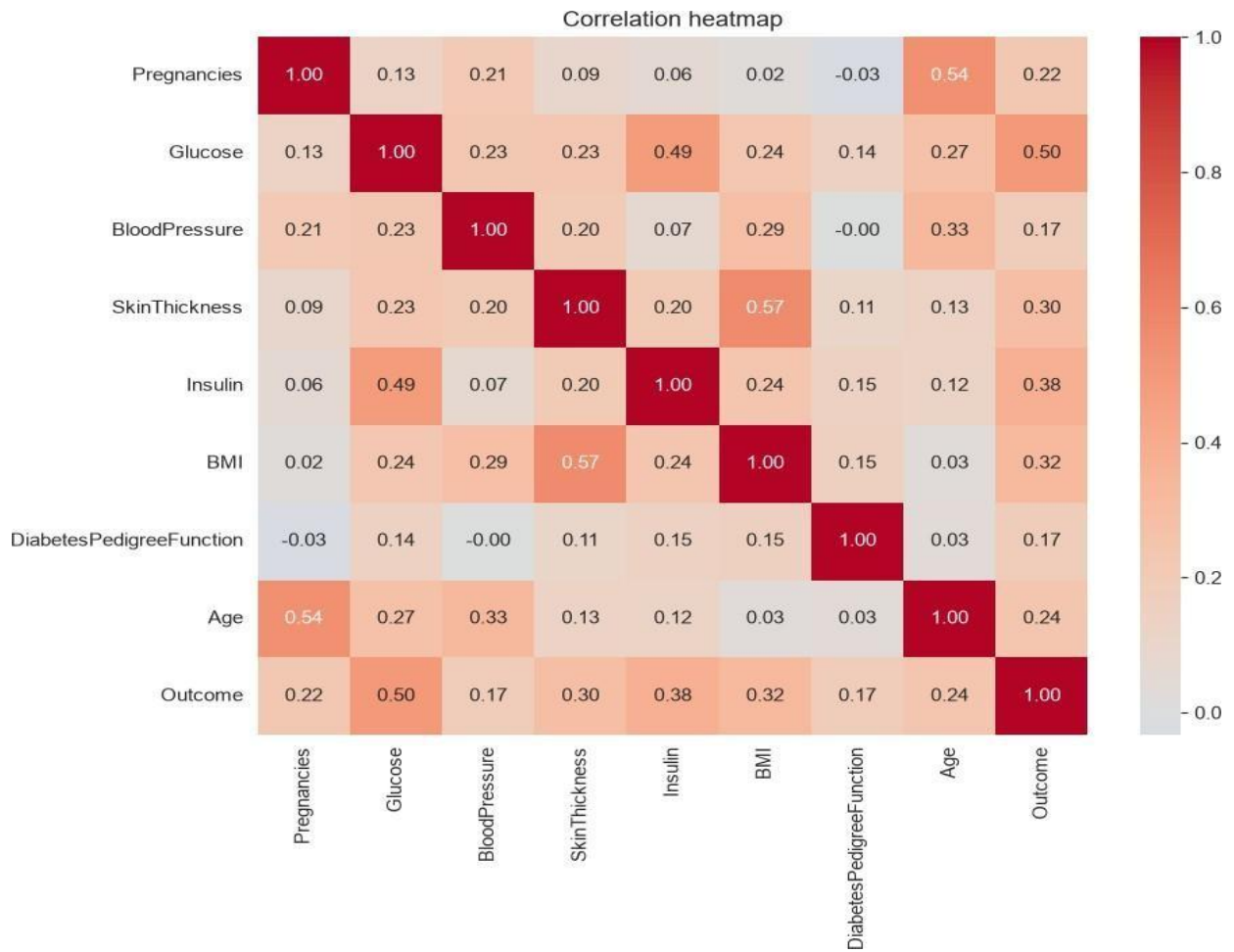
### Histograms



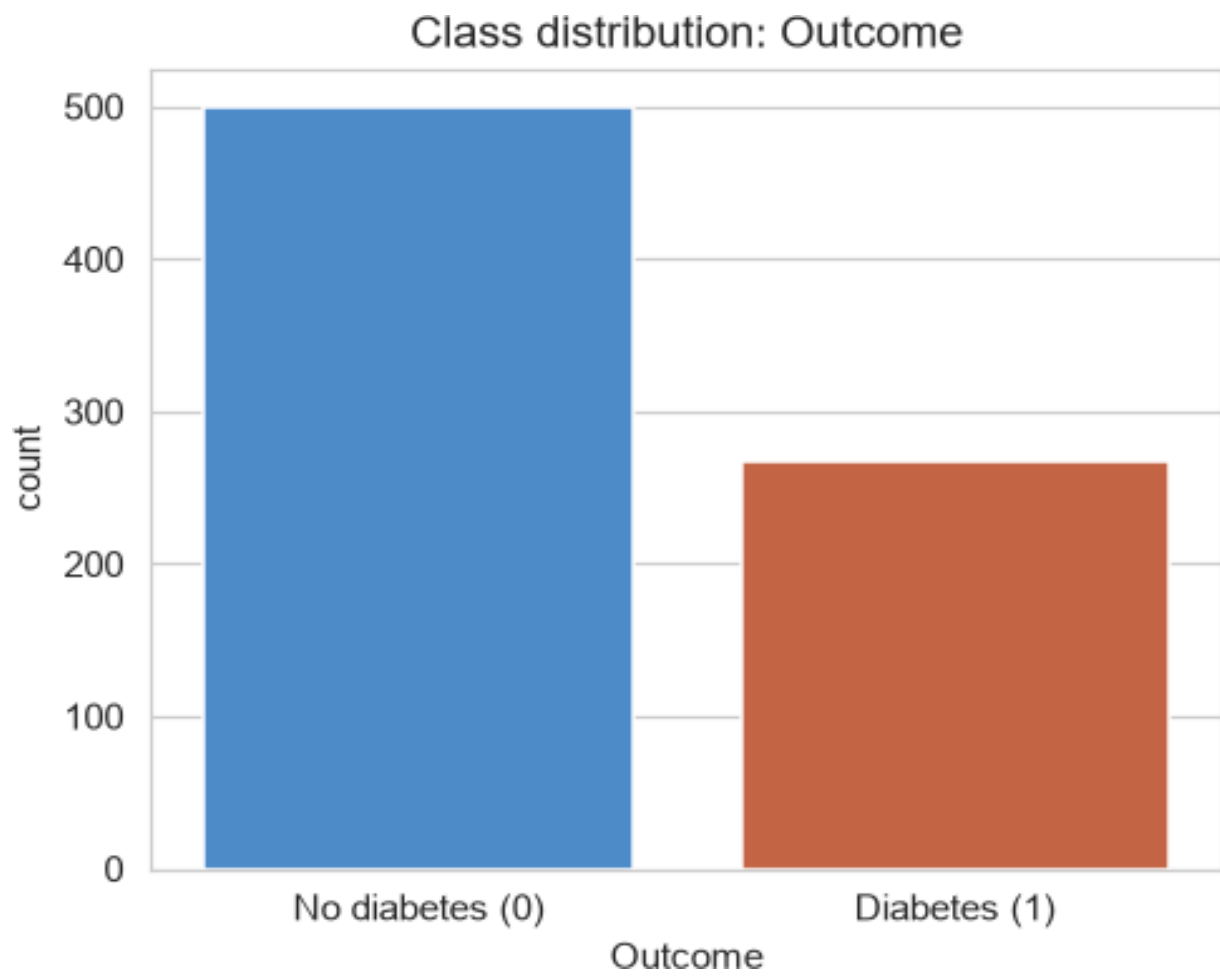
## Scatter: Glucose vs BMI



# Correlation heatmap



## Class Distribution

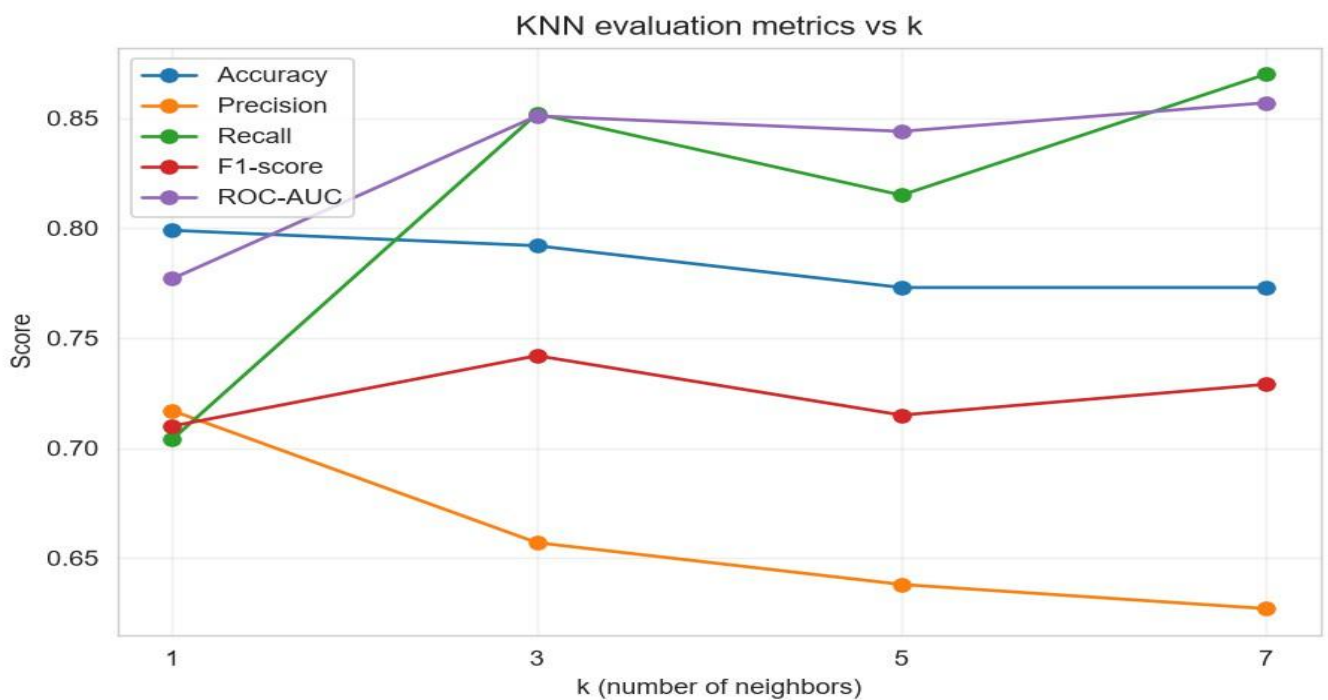


### 3. Model Results (*KNN vs Naive Bayes*)

| Model       | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|-------------|----------|-----------|--------|----------|---------|
| KNN         | 0.773    | 0.638     | 0.815  | 0.715    | 0.844   |
| Naive Bayes | 0.734    | 0.6       | 0.722  | 0.655    | 0.813   |

### 4. KNN performance across k values

| k | Model     | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|-----------|----------|-----------|--------|----------|---------|
| 1 | KNN (k=1) | 0.799    | 0.717     | 0.704  | 0.71     | 0.777   |
| 3 | KNN (k=3) | 0.792    | 0.657     | 0.852  | 0.742    | 0.851   |
| 5 | KNN (k=5) | 0.773    | 0.638     | 0.815  | 0.715    | 0.844   |
| 7 | KNN (k=7) | 0.773    | 0.627     | 0.87   | 0.729    | 0.857   |



## 5. Best Model Comparison

### Best-tuned KNN vs Naive Bayes

| Model       | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|-------------|----------|-----------|--------|----------|---------|
| KNN (k=3)   | 0.792    | 0.657     | 0.852  | 0.742    | 0.851   |
| Naive Bayes | 0.734    | 0.6       | 0.722  | 0.655    | 0.813   |

